Database Systems Project

TechNexus IT Services – Database Design & Query Project

Using Oracle APEX (SQL)

Course: Database Management Systems
Term: Spring 2026

Project Requirements

This database project is a group assignment. Use your Oracle APEX account to create a database for TechNexus IT Services – a mid-sized managed IT services company providing software licensing, technical support, and cloud solutions to business clients.

Due Date (complete project submission): 1:00pm Tuesday, April 28th 2026

Deliverables:

1) Tables created in your schema (One database per group; Email the instructor regarding whose APEX account your group is using).
2) ONE script including all SQLs in your group account (DDL, DML, and DCL).
   a. Provide comments with question # indicating what each SQL does. Use comment devices (two hyphens, '--'). Otherwise, you may not receive a full/partial grade.
   e.g. - - Step 3 Q2: Count employees per department)
3) Run your script file and confirm all SQLs execute without errors (check output file after execution). The number of errors in the 'Results' file will be negatively reflected in the project grade.
4) Extra credits for professional formatting using ORDER BY and appropriate ALIAS for column and table names.

Project Guidelines

Case Background

TechNexus IT Services is a growing managed IT services company that provides software solutions, technical support contracts, and cloud infrastructure to corporate clients. TechNexus needs a relational database system to manage its clients, employees, IT assets, software licenses, support tickets, projects, service contracts, and invoices.

Database Tables Overview

The TechNexus DB consists of 8 tables:

Table 1. TechNexus DB Tables and Relationships

Table Name	Description
CLIENT	Corporate clients who purchase services from TechNexus
EMPLOYEE	TechNexus staff members who handle support and projects - Each employee may be assigned to zero or more support tickets (optional many). - Each ticket must be assigned to exactly one employee (mandatory one). - Each employee may lead zero or more projects (optional many). - Each project must have exactly one lead employee assigned (mandatory one).
DEPARTMENT	Internal departments within TechNexus - Each department must have one or more employees (mandatory many). - Each employee must belong to exactly one department (mandatory one).
SOFTWARE	Software products that TechNexus licenses to clients - Each software product may appear in zero or more license records (optional many). - Each license record must reference exactly one software product (mandatory one).
LICENSE	Individual software license records issued to clients - Each client may hold zero or more software licenses (optional many). - Each license record must belong to exactly one client (mandatory one).

TICKET	IT support tickets submitted by clients - Each client may have submitted zero or more support tickets (optional many). - Each ticket must be associated with exactly one client (mandatory one).
PROJECT	IT consulting/implementation projects for clients - Each client may have zero or more projects commissioned (optional many). - Each project must be associated with exactly one client (mandatory one).
INVOICE	Billing records for services rendered to clients - Each client may have zero or more invoices issued (optional many). - Each invoice must be linked to exactly one client (mandatory one).

Project Guidelines

Step 1: ERD (By 4/14)

- Draw an ERD showing all entities and relationships among the relations of the TechNexus DB on the **description** in Table 1.
- Ensure that each relationship in the database is clearly represented with both cardinality (One-to-One, One-to-Many, or Many-to-Many) and optionality (mandatory or optional) using Visio notations in the lecture note.
- Visio is available for download from the URI myApps. Submissions created with other software will not be graded.

Step 2: Create the Database (DDL) (By 4/14)

Create all 8 tables in your Oracle APEX schema. The following constraints must be implemented:

- Create 8 tables: CLIENT, EMPLOYEE, DEPARTMENT, SOFTWARE, LICENSE, TICKET, PROJECT, and INVOICE.
- Referential integrity constraints using PRIMARY KEY and FOREIGN KEY must be implemented when creating the tables.
- All attributes do NOT allow NULL values unless explicitly stated.
- Be creative and thoughtful with data types (VARCHAR2, NUMBER, DATE, CHAR, etc.).
- Create tables in the correct order to satisfy foreign key dependencies (parent tables first).

Table Schemas

1. DEPARTMENT Table

Column Name	Data Type	Constraint	Description
DEPT_ID	NUMBER(4)	PRIMARY KEY	Unique department identifier
DEPT_NAME	VARCHAR2(50)	NOT NULL, UNIQUE	Department name (e.g., 'Cloud Services')
DEPT_LOCATION	VARCHAR2(20)	NOT NULL	City - Physical or virtual location of department
DEPT_STATE	VARCHAR2(2)	NOT NULL	State - Physical or virtual location of department
DEPT_BUDGET	NUMBER(10,2)	NOT NULL	Annual budget allocated to department

2. CLIENT Table

Column Name	Data Type	Constraint	Description
CLIENT_ID	NUMBER(6)	PRIMARY KEY	Unique client identifier
CLIENT_NAME	VARCHAR2(100)	NOT NULL	Company/organization name
CLIENT_INDUSTRY	VARCHAR2(50)	NOT NULL	Industry sector (e.g., 'Healthcare')
CLIENT_EMAIL	VARCHAR2(100)	NOT NULL, UNIQUE	Primary contact email
CLIENT_PHONE	CHAR(12)	NOT NULL	Contact phone (XXX-XXX-XXXX)
CLIENT_STATE	CHAR(2)	NOT NULL	State abbreviation
CLIENT_JOIN_DATE	DATE	NOT NULL	Date client signed up
CLIENT_TIER	VARCHAR2(20)	NOT NULL	Service tier: Basic / Pro / Enterprise

3. EMPLOYEE Table

Column Name	Data Type	Constraint	Description
EMP_ID	NUMBER(6)	PRIMARY KEY	Unique employee identifier
EMP_FNAME	VARCHAR2(50)	NOT NULL	First name
EMP_LNAME	VARCHAR2(50)	NOT NULL	Last name
EMP_EMAIL	VARCHAR2(100)	NOT NULL, UNIQUE	Company email address
EMP_ROLE	VARCHAR2(50)	NOT NULL	Job role (e.g., 'Systems Engineer')
EMP_HIRE_DATE	DATE	NOT NULL	Date of hire
EMP_SALARY	NUMBER(10,2)	NOT NULL	Annual salary
EMP_CERT	VARCHAR2(100)	NULL	Primary certification (e.g., 'AWS-SAA')
DEPT_ID	NUMBER(4)	FK → DEPARTMENT	Department the employee belongs to

4. SOFTWARE Table

Column Name	Data Type	Constraint	Description
SW_ID	VARCHAR2(10)	PRIMARY KEY	Software product code (e.g., 'SW-MS365')
SW_NAME	VARCHAR2(100)	NOT NULL	Software product name
SW_VENDOR	VARCHAR2(50)	NOT NULL	Publisher / vendor name
SW_CATEGORY	VARCHAR2(50)	NOT NULL	Category: Security / Productivity / Cloud / etc.
SW_VERSION	VARCHAR2(20)	NOT NULL	Current version string
SW_UNIT_COST	NUMBER(8,2)	NOT NULL	Cost per license seat to TechNexus
SW_UNIT_PRICE	NUMBER(8,2)	NOT NULL	Price per license seat billed to clients

5. LICENSE Table

Column Name	Data Type	Constraint	Description
LIC_ID	NUMBER(8)	PRIMARY KEY	Unique license record identifier
CLIENT_ID	NUMBER(6)	FK → CLIENT	Client who holds the license
SW_ID	VARCHAR2(10)	FK → SOFTWARE	Licensed software product
LIC_SEATS	NUMBER(5)	NOT NULL	Number of licensed seats
LIC_START_DATE	DATE	NOT NULL	License activation date
LIC_END_DATE	DATE	NOT NULL	License expiration date

6. TICKET Table

Column Name	Data Type	Constraint	Description
TICKET_ID	NUMBER(8)	PRIMARY KEY	Unique support ticket identifier
CLIENT_ID	NUMBER(6)	FK → CLIENT	Client who submitted the ticket
EMP_ID	NUMBER(6)	FK → EMPLOYEE	Employee assigned to the ticket
TICKET_DATE	DATE	NOT NULL	Date ticket was created
TICKET_CATEGORY	VARCHAR2(50)	NOT NULL	Type: Network / Security / Software / Hardware
TICKET_PRIORITY	VARCHAR2(10)	NOT NULL	Priority: Low / Medium / High / Critical
TICKET_STATUS	VARCHAR2(20)	NOT NULL	Status: Open / In Progress / Resolved / Closed
TICKET_RESOLVE_DATE	DATE	NULL	Date ticket was resolved (NULL if open)

7. PROJECT Table

Column Name	Data Type	Constraint	Description
PROJ_ID	NUMBER(6)	PRIMARY KEY	Unique project identifier
CLIENT_ID	NUMBER(6)	FK → CLIENT	Client the project is for
EMP_ID	NUMBER(6)	FK → EMPLOYEE	Lead employee / project manager
PROJ_NAME	VARCHAR2(100)	NOT NULL	Project title
PROJ_START_DATE	DATE	NOT NULL	Project kick-off date
PROJ_END_DATE	DATE	NULL	Expected or actual completion date
PROJ_BUDGET	NUMBER(10,2)	NOT NULL	Approved project budget

8. INVOICE Table

Column Name	Data Type	Constraint	Description
INV_ID	NUMBER(8)	PRIMARY KEY	Unique invoice identifier
CLIENT_ID	NUMBER(6)	FK → CLIENT	Client being billed
INV_DATE	DATE	NOT NULL	Date invoice was issued
INV_DUE_DATE	DATE	NOT NULL	Payment due date
INV_AMOUNT	NUMBER(10,2)	NOT NULL	Total invoice amount
INV_STATUS	VARCHAR2(20)	NOT NULL	Payment status: Pending / Paid / Overdue

Step 3: Populate the Database (DML – INSERT) (By 4/21)

Populate all 8 tables according to the requirements below.

Table	Min Records	Data Requirements
DEPARTMENT	5 records	Must include departments: Cloud Services, Cybersecurity, Technical Support, Software Licensing, and IT Consulting.
CLIENT	20 records	Clients must come from at least 6 different states including TX and CA. Must include client ID 10001 (TechPeak Corp, Austin TX, Enterprise tier) and 10002 (Greenfield Health, Boston MA, Pro tier).

EMPLOYEE	15 records	At least 2 employees per department. Include at least 3 employees with a certification (EMP_CERT).
SOFTWARE	10 records	SW_CATEGORY must include at least 4 types: Security, Productivity, Cloud, and Collaboration. Include SW-MS365 (Microsoft 365) and SW-CRWD (CrowdStrike Falcon).
LICENSE	20 records	At least 15% of licenses must be expired (LIC_END_DATE < SYSDATE). Some clients may hold licenses for multiple software products.
TICKET	25 records	At least 20% of tickets must remain unresolved (TICKET_RESOLVE_DATE IS NULL). Include all four TICKET_PRIORITY values. Include all four TICKET_CATEGORY values.
PROJECT	12 records	At least 2 projects must have no end date (ongoing). PROJ_BUDGET must range from $5,000 to $250,000.
INVOICE	20 records	At least 25% of invoices must have INV_STATUS = 'Overdue'. Each client must have at least one invoice.

Sample data (required for all group databases)

Table	Data Requirements
DEPARTMENT	(1101, Cloud Services, Boston, MA, 850000) (1102, Cybersecurity, Austin, TX, 920000)
CLIENT	(110001, TechPeak Corp, Technology, contact@techpeak.com, 512-555-0101, TX, 2020-03-15, Enterprise) (110002, Greenfield Health, Healthcare,info@greenfield.com, 617-555-0202, MA, 2021-07-01, Pro)
EMPLOYEE	(110001, Sarah, Nguyen, s.nguyen@technexus.com, Cloud Architect, 2019-04-10, 112000, AWS-SAA ,10) (110002, Marcus, Webb, m.webb@technexus.com, Security Analyst, '2020-09-01, 98000, CISSP ,20)
SOFTWARE	(SW-MS365, Microsoft 365, Microsoft, Productivity, 2024.1, 8.00, 15.00) (SW-CRWD, CrowdStrike Falcon, CrowdStrike, Security, 6.5, 18.50, 35.00)
LICENSE	(11000001, 110001, SW-MS365, 50, 2023-01-15, 2026-01-15) (11000002, 110001, SW-CRWD, 50, 2023-01-15, 2026-01-15);
TICKET	(110001, 110001, 110001, 2024-01-05, Software, High, Resolved, 2024-01-12) (110002, 110002, 110002, 2024-01-10, Security, Critical, Resolved, 2024-01-11)
PROJECT	(110001, 110001, 110001, Cloud Migration Phase 1, 2023-03-01, 2023-09-30, 120000) (110002, 110002, 110002, Cybersecurity Audit, 2023-05-15, 2023-08-15, 45000);
INVOICE	(11000001, 110001, 2024-10-01, 2024-10-31, 47500.00, Paid) (11000002, 110002, 2024-11-15, 2024-12-15, 12300.00, Overdue)

Step 4: Data Import & Grant Permissions to Partner Group (DCL) (By 4/21)

1) Find a partner group, and grant a SELECT permission to access your tables and let them import data from all tables using the following command:

   "GRANT SELECT ON ***YOUR_TABLE*** TO ***PARTNER_SCHEMA***;"

2) Request a permission to access data in the tables of your partner group.
3) Import all data from your partner group tables.
   a. Select and import only records that do not exist in your DB from your partner's all tables.
      - Create a DML including “INSERT and SELECT” so that you can import records into your schema while retrieving them from the other schema at the same time. Use a non-correlated subquery to import only data that does not exist in your table.
      - If the PK is a composite one, you may need to use 1) a correlated subquery or 2) concatenation operator, ||, to create a string instead of individual columns.

b. You need to use either correlated (if the primary key is composite) or non-correlated subqueries (if the primary key is single).

Step 5: Write SELECT Queries (By 4/28)

Write SQL SELECT statements to answer the following questions. All query results must include ORDER BY clauses and use ALIAS names to format column headers professionally.

Q1. Show the number of records in each of the 8 tables in a single query using union joins.

Q2. How many clients are based in Texas (TX)?

Q3. List all states and the number of clients in each state using Vector Aggregate (GROUP BY).

Q4. Write a single SQL command to increase the unit price (SW_UNIT_PRICE) of all software products in the 'Security' category by $5.00.

Q5. Display the software ID, software name, vendor, and unit price for all software products where the unit price is greater than $20.00.

Q6. Write a query to display client name, software name, software category, number of license seats, and the total license cost (seats × unit price) for all active licenses (LIC_END_DATE >= SYSDATE).

Q7. Display software category and the average unit price for software in each category.

Q8. Display client name, client tier, department name, and the total number of open tickets for clients in the 'Enterprise' or 'Pro' service tier. Include only clients with at least 1 open ticket.

Q9. List the client ID, client name, and state for all clients that do NOT have any invoice on record. Use a subquery.

Q10. Display the ticket ID, client name, assigned employee full name, ticket category, priority, open date, and the number of days the ticket has been open for all tickets that are NOT yet resolved or closed. Sort by priority and days open.

Q11. Display the client ID, client name, and the total invoice amount billed to each client. Include all clients, even those with no invoices (show 0 for clients with no invoices).

Q12. Display the client name and total amount billed for the client who has the highest total invoice amount.

Q13. Which software product generates the most total license revenue for TechNexus? Display the software name, vendor, and total revenue (seats × unit price across all licenses).

Q14. Display the software ID, software category, average unit price of software in that category, the unit price of each individual software product, and the percentage difference between each product's price and its category average. (Format as shown below.)

Note: The percentage difference = ((Unit Price − Avg Category Price) / Avg Category Price) × 100. A positive value means the product is priced above its category average; a negative value means below.

Q15. Display the employee full name, department name, ticket ID, ticket category, ticket priority, client name, and the number of days it took to resolve the ticket for all tickets that have been resolved. Only include tickets where the resolution took more than 3 days. Sort the results by the number of days to resolve in descending order, then by employee last name.

Grading Rubric

Component	Points	Criteria
Step 1 – ERD	15 pts	Correct ERD including tables, PKs, FKs, cardinality, optionality.
Step 2 – DDL (Table Creation)	15 pts	All 8 tables created with correct PK/FK constraints, data types, and NOT NULL rules
Step 3 – DML (Data Population)	15 pts	All tables meet minimum record counts; required specific records included; data variety
Step 4 – DCL (Grant & Import)	15 pts	Correct GRANT statements; non-duplicate import using subquery
Step 5 – SELECT Queries (15 questions)	37 pts	Correct logic, results, ORDER BY, and ALIAS for all queries
Script Formatting & Comments	3 pts	Proper -- comment labels on every SQL block; readable formatting

Good luck! For questions about Oracle APEX setup, refer to the Oracle APEX lecture notes on the course portal.